From Numerical Search to Checkable Mathematics

A mathematician's guide to the experiment-craft ecosystem

experiment-craft documentation

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1 What this guide lets you do

Use this software to turn a mathematical question into a reproducible computation: formulate an experiment, search for candidates, run a checker, and retain the evidence and failures. An AI agent can write and operate the experiment, but the mathematical claim must come from a stated argument or checker with explicit assumptions.

This guide was checked on 27 October 2026 against experiment-craft `63235813cbd2`, which includes the reviewed dashboard and recorded certificate views, and against the upstream revisions listed at the end. The native revisions are frozen by `baseline-lock.json`. The commands were run on one Linux machine with local nodes; separate physical machines and a physical phone were not tested.

Three useful starting points are:

- **A working mathematical pilot:** tune a proposed approximate inverse, then validate and replay a scalar or PDE point certificate.
- **A solver experiment:** compare bounded black-box searches, solve a small integer model, or check a smooth constrained first-order point.
- **Your own exact computation:** run a transparent finite producer and checker as trusted commands. The Ramsey design below shows how.

A research campaign on chains through derivative roots shows these pieces at work on an open problem: 96 native jobs, 14 exact finite certificates, two written obstructions to proposed lemmas, and the failures along the way.

1.1 The distinctions to keep

Observation	What it supports
Small floating residual or good objective	A numerical candidate worth examining
Feasible point, numerical bound/gap, or KKT residual	The native solver's numerical claim under its stated tolerances
A successful theorem-specific interval validation and fresh replay	The stated local mathematical conclusion, under the verifier's trusted assumptions
A sealed output or a hash match	Identity of bytes, not validity of the mathematics
Dashboard `reported_accepted`	The admitted application reports acceptance consistently with its execution record; the dashboard ran no checker

How convincing a picture looks does not decide the strongest claim. In particular, an inconclusive validation is not a proof of nonexistence, and a small discretized PDE residual does not control an unresolved infinite tail.

2 Which repository does what?

Repository	Mathematical role	Use it when
experiment-craft	Campaigns, dependent steps, budgets, immutable inputs, reports and local dashboard	You need a reproducible collection of computations and their evidence

Repository	Mathematical role	Use it when
optimization-jobs	Executes optimization jobs and trusted commands; owns admission, files, handles, cancellation and recovery; supplies CLI/MCP	You or an agent must run and inspect a bounded attempt
fcmaes-rust	Native Rust black-box, multi-objective and quality-diversity search; application workers; the validated-numerics tutorial	Your objective lives in Rust, or you use its prepared mathematical applications
fast-cma-es	Python `fcmaes` with a C++ backend: numerical search, parallel retry, Pareto and diversity methods	Your objective is a Python function; run it directly or as a trusted command
LP-IP-craft	Linear and mixed-integer linear models in Rust, with Python bindings and a worker	The objective and rows are linear; some variables may be integer
SQP-craft	Smooth constrained nonlinear optimization with derivatives, Python bindings and a worker	A local first-order solution is useful; the current dense pilot supports at most 256 variables
pykep-core-rust	Astrodynamics: propagation, Lambert arcs, ephemerides, sensitivities and low-thrust legs	Your mathematical objective is a trajectory model; this supplies physics rather than an optimizer or proof checker

The two fcmaes repositories are alternative implementations, not a mandatory pair. The Python `fcmaes` package is not the Rust job worker. The runtime has no optimizer of its own, and the orchestrator does not replace runtime supervision or file-transfer verification.

For the CAP pilot, the theorem-specific Julia application is under the validated-numerics tutorial. Its RadiiPolynomial and IntervalArithmetic environment is separate from the Rust optimizer core. This is the source of the rigorous calculations. `experiment-craft` only coordinates its commands and reads the resulting records.

2.1 Pick an interface

A direct library call is enough for a short interactive calculation. Use a job when you need a handle, limits, staged inputs and later inspection. Use a campaign for seed/start sweeps, preparation dependencies or a chain such as search, audit and acceptance. Use the dashboard to operate an already initialized campaign and read its saved observations.

What a stock worker accepts depends on the worker. The LP-IP worker solves any LP or MIP given as a JSON model file, which is data. The fcmaes and SQP workers solve only models registered in their native code; they do not accept a new equation because an agent writes its name in JSON. For a new nonlinear model, implement an application worker, use the library in your own program, or stage that program as an explicitly enabled trusted command.

3 Application areas and actual coverage

Area	Available route	Present boundary
Validated roots and nonlinear steady states	The two fixed CAP applications, fresh verification/replay and all-arm audit	Positive cube root and one order-32 Cahn-Hilliard case only; no general theorem loader
Counterexample and extremal search	Black-box optimization of a violation or extremal score, followed by an independent checker	No universal counterexample decoder or proof extraction
Finite combinatorics	Trusted enumeration/search/check commands, as in the Ramsey design below	Completeness is the application's responsibility
LP, integer selection, allocation and discrete designs	LP-IP-craft with explicit rows, finite bounds on every variable of a MIP, and independent model checks	Numerical solver claims; no automatic exact certificate for arbitrary MIPs
Smooth inequalities, constrained profiles and parameter fits	SQP-craft or a numerical library with gradient/Jacobian checks and multistart	Local first-order evidence; no global-optimality or nonlinear infeasibility proof
Pareto exploration and diverse constructions	fcmaes MODE and MAP-Elites/library diversity tools	A population/archive is not an exhaustive classification or a certified Pareto set
Expensive simulation calibration	The experimental surrogate profile of fcmaes-rust, or ordinary search/retry	Surrogate profile: scalar, continuous, deterministic boxes; up to 32 variables and 256 observations in total, including warm data
Dynamics, trajectories and astrodynamics	pykep-core-rust models with fcmaes or smooth optimization	Check units, ephemerides, discretization and model validity separately
Fluid models and inverse design	Native simulation tutorials or a custom trusted simulator/checker	The room-ventilation tutorial is an educational 2D lattice-Boltzmann simulation, not engineering CFD and not a rigorous Navier-Stokes or Euler proof

These are different levels of readiness. A method being useful in an area does not mean a theorem checker for that entire area exists. New singularity proofs, branch continuation, stability proofs, parameter-region coverage and semantic replay across environments require additional mathematics and code.

A good custom proof-search design optimizes quantities that make a fixed verification argument easier: an approximate inverse, a weight, a basis or a bound allocation. Keep the equation, exact target, admissible function class and conclusion fixed while comparing methods. Changing them is a new problem, not an improved result for the old one.

4 First setup and a seven-job tour

The supported workflow here uses a Linux source checkout. Use Python 3.12 or newer, Git, a C/C++ build toolchain and rustup. The native build toolchain is Rust 1.88.0. Start from experiment-craft's recorded source rather than assuming that a similarly numbered package release contains every tutorial feature.

```
git clone https://github.com/dietmarwo/experiment-craft.git
cd experiment-craft
python3.12 -m venv .venv
.venv/bin/python -m pip install -r requirements-baseline.txt
.venv/bin/python -m pip install -e .
rustup toolchain install 1.88.0 --profile minimal
.venv/bin/python scripts/baseline.py fetch build smoke \
  --workspace .baseline --jobs 2
```

`fetch` obtains and verifies the four exact native source revisions and their shared API compatibility. `build` stages the runner, its MCP server and the three workers; `smoke` runs real jobs. The baseline's `bin/manifest.json` records binary hashes. Every action verifies the sources first, and `smoke` also checks the staged binary hashes. Building can take several minutes and needs network access to GitHub, crates.io and PyPI. The full upstream checks (`check`) additionally need the latest stable Rust and the OpenSSH client; see the README.

Run the existing tour with a fresh output directory:

```
.venv/bin/python examples/campaigns/local-mixed/run.py \
  --bin-dir .baseline/bin --output "$PWD/guide-runs/mixed"
```

It explicitly creates a one-slot local node that allows trusted commands and registers the three workers. Seven native jobs should complete: a command that stages an LP model, a three-seed fcmaes search, the LP/IP and SQP solves, and a command that checks their results. This is setup by an example program, not automatic node administration by the client.

The LP/IP example is exactly

$$\min\ 5 + 2x_1 + 3x_2, \qquad x_1 + x_2 = 1, \qquad x_1, x_2 \in \{0, 1\}.$$

Its optimum is 7 at $(1, 0)$. The worker reports the claim `numerical_optimum`: the native status `optimal`, after the worker's own recheck of objective, bounds, rows and integrality. The smooth example minimizes

$$\sum_{i=1}^{4} x_i^2, \qquad \sum_{i=1}^{4} x_i = 1, \qquad 0 \le x_i \le 1.$$

Its known answer is $x_i = 1/4$ with objective $1/4$. The worker reports `first_order`: native convergence plus its own KKT recheck. The example's check step requires these two claims and compares the reported objectives with 7 and $1/4$; it does not recompute the points. The elementary global argument for this special model is not a guarantee supplied by SQP. Both examples are small plumbing checks, not performance comparisons with mature solvers.

5 Worked CAP example: a positive cube root

The problem is the fixed equation

$$F(u) = u^3 - 2 = 0, \qquad u > 0, \qquad \varepsilon = \frac{1}{10^{10}}.$$

The coefficient is an exact rational and the physical error target is exact. Candidate coefficients and approximate inverse entries are saved as lossless binary64 bit patterns. Keep those original bytes; rounding a printed value creates a different candidate. The floating approximation is a proposed center $\bar{u}$, not an interval proof. A candidate also supplies a proposed scalar inverse A.

5.1 What optimization changes

The reference coefficients come from a floating Newton solve. The search changes a multiplier on the approximate inverse within $[0.5, 1.5]$. It does not optimize the equation, accuracy target or theorem. The frozen protocol has nine arms: reference, grid, local pattern search, and random/CMA-ES/BiteOpt with seeds 17 and 29.

There are at most 25 search calls per non-reference arm; the reference uses one. Including nine winner rechecks and one warm-up gives 211 floating estimates per study. The score estimates a verification safety margin, not wall-clock time. A favorable score is only a proposal for the rigorous step.

For intuition, the fixed-point map is $T(u) = u - AF(u)$; the verifier checks $A \neq 0$, so a fixed point of T is a root of F. It sets the radius to $r = 2 \sup Y$ and bounds, with interval arithmetic,

$$Y \geq |AF(\bar{u})|, \qquad Z_1 \geq \sup_{u \in B(\bar{u}, r)} |1 - A\,DF(u)|.$$

It then checks

$$Y + (Z_1 - 1)r \leq 0, \qquad Z_1 < 1, \qquad 0 < r \leq \varepsilon,$$

and that the whole ball is positive, which selects the positive root. Together these give existence and uniqueness in that ball. The complete contract is THEOREM.md. The software does not claim a maximal uniqueness radius.

5.2 Prepare the Julia and search environments

Use **Julia 1.10.10**. Set `GUIDE_JULIA` below to its absolute executable; it is an explicit installation choice. Do not run `Pkg.update()` when reproducing this study. From experiment-craft's root:

```
export GUIDE_JULIA=/absolute/julia-1.10.10/bin/julia
export GUIDE_APP="$PWD/.baseline/repos/fcmaes-rust/tutorials/validated-numerics"
export JULIA_DEPOT_PATH="$PWD/guide-julia-depot"
export JULIA_NUM_THREADS=1 OPENBLAS_NUM_THREADS=1
export OMP_NUM_THREADS=1 MKL_NUM_THREADS=1
export JULIA_PKG_USE_CLI_GIT=true
"$GUIDE_JULIA" --startup-file=no --project="$GUIDE_APP/julia" \
  -e 'using Pkg; Pkg.instantiate()'
cargo +1.88.0 build --release --locked \
  --manifest-path "$GUIDE_APP/search/Cargo.toml" \
  --target-dir "$PWD/guide-search-target"
```

The Julia project/manifest pins RadiiPolynomial 0.11.10 and IntervalArithmetic 1.0.12, including their source identities and dependencies. The instantiated depot and interpreter must exist on the node; the job does not transfer their installations. Keep that trusted environment controlled.

5.3 Run, inspect and retain

```
.venv/bin/python examples/campaigns/certificates/run.py \
  --fcmaes-source .baseline/repos/fcmaes-rust \
  --binary guide-search-target/release/cap-search \
  --julia "$GUIDE_JULIA" --depot "$JULIA_DEPOT_PATH" \
  --runner .baseline/bin/optimization-run \
  --problem scalar-cuberoot --output "$PWD/guide-runs/scalar"
```

This wrapper explicitly creates its own node. It runs three admitted jobs: search, an audit that verifies and replays every final arm, and strict acceptance. When every verification completes, the audit starts 18 fresh Julia processes, one after another inside its one-slot attempt; a replay is skipped when its verification produced no usable result. The wrapper retains the receipt, writes an offline report, rechecks the nine certificates' bounds in exact rational arithmetic and confirms that continuation adds no jobs. Use new directories for a new scientific attempt.

The recorded runs reported all nine arms accepted for both problems, and CI repeats both campaigns on every pull request. This is evidence that these easy candidates validate, not evidence that one optimizer is better. The report is `verification/verification.json` inside `guide-runs/scalar/`; the workspace is `guide-runs/scalar/campaign`.

For manual collection and offline reporting:

```
.venv/bin/experiment-craft artifacts \
  --workspace guide-runs/scalar/campaign
.venv/bin/experiment-craft collect-artifact \
  --workspace guide-runs/scalar/campaign \
  --step accept --output acceptance
.venv/bin/experiment-craft verification-report \
  --workspace guide-runs/scalar/campaign --step accept \
  --destination "$PWD/guide-runs/scalar-report"
```

The report destination is outside the campaign and native node. Repeated collection checks a retained copy offline; repeated reporting runs no checker and starts no new job. Fresh scientific replay is a new explicit computation, not a call to `continue`.

6 Worked PDE example: a Neumann steady state

The second fixed problem is

$$u'' + \beta(u - u^3 - c) = 0 \quad \text{on } [0,1],$$
$$u'(0) = u'(1) = 0, \qquad c = 0, \qquad \beta = \frac{1285}{32}.$$

The verifier works in the real, even, 2-periodic function space, with

$$u(x) = a_0 + 2\sum_{k\geq 1} a_k \cos(k\pi x), \qquad \|a\|_X = |a_0| + 2\sum_{k\geq 1} |a_k|.$$

Here the factor two is essential. Since $\|u\|_\infty \leq \|a\|_X$, an accepted coefficient-norm radius $r \leq 10^{-10}$ controls the same physical sup-norm error. The first cosine mode is certified positive, proving the solution is nonconstant and fixing its sign orientation. Uniqueness is local in this stated even function space. There is no stability or full-branch conclusion.

The saved approximation has modes through $K = 32$ and a finite approximate inverse. The verifier extends the inverse by an inverse-Laplacian tail and bounds both the finite and the infinite-tail derivative defects. The tail bound is

$$Z_{1,\mathrm{tail}} = \frac{\beta(1 + 3\|\bar{u}\|_X^2)}{[\pi(K+2)]^2}.$$

With the radius set to $r = 10 \sup Y$, the second-order conditions are

$$Y + (Z_1 - 1)r + \tfrac{1}{2} Z_2 r^2 \le 0, \qquad Z_1 + Z_2 r < 1, \qquad 0 < r \le 10^{-10}.$$

These formulas summarize this particular verified argument; they are not a universal theorem obtained by substituting arbitrary residuals. The verifier also checks the injectivity of the finite inverse, guaranteed arithmetic, the branch condition and the analytic tail bound. The function-space argument itself is part of the trusted, theorem-specific adapter. Read the mathematical contract before extending it.

Search tunes two inverse-row multipliers in $[0.5, 1.5]^2$: one for the zero mode and one shared by nonzero rows. The reference coefficient vector, order, exact target and tail stay fixed. To run this example, repeat the certificate command above with `--problem cahn-hilliard` and output `"$PWD/guide-runs/pde"`. The recorded run reported all nine arms accepted, and the exact rational recheck passed for each. It is a reproduction of a known reference validation, not a new PDE theorem or a blowup construction.

7 Your own finite check: Ramsey witnesses

An exact finite statement fits the same pattern with two trusted commands of your own: a producer that writes a certificate and a separate checker that reads it. This repository does not ship such an application; the following design uses the known statement

$$R(3, 3) = 6.$$

Here $R(3, 3)$ is the least number of vertices forcing a triangle or an independent triple in every simple graph. Every labeled simple graph on six vertices has 15 possible edges, hence $2^{15} = 32{,}768$ edge masks. The producer writes one monochromatic triple for every mask: a triple with all three edges present is a triangle; all three absent is an independent triple. The checker reads exactly one witness per mask and checks its three edge bits with a separately coded edge-index formula. It also checks that the five-cycle has neither kind of triple, which gives the lower example. Coverage is then explicit rather than inferred from random sampling or from an optimizer finding nothing.

In the campaign, the producer declares its certificate as a required output, and the checker's input selects that output, which binds the checker to the exact bytes the producer sealed:

```
{"steps": [
  {"id": "enumerate", "job": {"schema_version": "execution.command/0.1",
    "argv": ["/usr/bin/python3", "enumerate.py"],
    "inputs": [{"path": "enumerate.py", "file": "enumerate.py"}],
    "outputs": [{"name": "witnesses", "path": "witnesses.bin",
                 "required": true, "max_bytes": 32768}]}},
  {"id": "verify", "job": {"schema_version": "execution.command/0.1",
    "argv": ["/usr/bin/python3", "verify.py"],
    "inputs": [{"path": "verify.py", "file": "verify.py"},
               {"path": "witnesses.bin", "from": {"step": "enumerate",
                "kind": "output", "output": "witnesses"}}],
    "outputs": [{"name": "verdict", "path": "verdict.json",
                 "required": true, "max_bytes": 4096}]}}
]}
```

Each job also needs the usual `env`, `resources`, `limits` and `replay` fields (campaign manifests). The checker's exit status decides whether its step succeeds; collect `verdict.json` with `collect-artifact` to read the conclusion. Such a result trusts the enumeration, the checker, Python's integer semantics and the execution environment; it is not a proof-assistant certificate. The built-in CAP audit view refuses a campaign of this shape: a general mathematical application does not automatically acquire a domain adapter.

8 A research campaign: chains through derivative roots

This case study reports a research campaign that used these tools on an open problem. It ran on 28 October 2026 as iterations 91–120 of an existing polynomial project, with a strategy review after every fifth step. Its results are an implemented search-and-check application, finite certificates, two written mathematical obstructions and a complete failure log; it does not claim a better asymptotic theorem. The campaign's code, results and evidence are kept in a separate archive, not in this repository; the numbers below are that archive's records, and the commands in the last subsection run from it.

8.1 The problem and its quantifiers

Let f have degree n, and let Z_m be the roots of the degree-m derivative row $f^{(n-m)}$, for $1 \le m \le n$. A chain chooses an actual root in every row:

$$z_m \in Z_m, \qquad L(z_n, \dots, z_1) = \sum_{m=2}^{n} |z_m - z_{m-1}|.$$

The campaign studies the shortest chain from the worst prescribed starting root,

$$B(f) = \max_{z_n \in Z_n} \min_{z_{n-1} \in Z_{n-1}, \dots, z_1 \in Z_1} \sum_{m=2}^{n} |z_m - z_{m-1}|.$$

The maximum over the starting root matters. A theorem asserting that *some* short chain exists has different quantifiers. For example, the report *A $\sqrt{n-1}$ bound for chains of zeros of successive derivatives* shows that for every polynomial with all roots in the closed unit disk, some chain has length at most $\sqrt{n-1}$; its chain does not start at a prescribed root, so it must not be read as a bound on $B(f)$.

The class here is also restricted. All roots lie in the closed unit disk, their centroid is zero, and **every nonconstant derivative has distinct roots, each a vertex of its convex hull**. A one-point row and a two-point row use the natural point/segment convention. Repeated-root constructions from the general problem are outside this strict class. Normalizing after translation can change the radius, so the exact checker verifies both centering and disk containment. Since the roots of every derivative share the centroid of the roots of f, $Z_1 = \{0\}$.

Dynamic programming computes the finite objective without guessing a chain:

$$V_1(0) = 0, \qquad V_m(z) = \min_{w \in Z_{m-1}} \left(|z - w| + V_{m-1}(w)\right), \qquad B(f) = \max_{z \in Z_n} V_n(z).$$

For a possible new proof, define

$$H_m = \operatorname{conv} Z_m, \qquad d_m = d_H(H_m, H_{m-1}), \qquad M_q(f) = \sum_{m=2}^{n} d_m^q.$$

The Gauss–Lucas theorem gives nested hulls, but nesting alone does not supply the stronger derivative-specific budget sought here. The campaign treats a large $B(f)$, a large $M_{4/3}(f)$ and the failure of a local lemma as three different objectives.

8.2 What had already improved

The preceding campaign, iterations 61–90, refined the argument of a strict-convexity manuscript dated 21 September 2026 for the upper bound of order $n^{1/3} \log n$. In the project this bound is called "Ingo's theorem", after Ingo Althöfer; the manuscript itself names no author. The comparison below is with that manuscript's argument, not a claim about priority in the literature.

Part of the proof	Manuscript	After iterations 61–90
Generic $\sum d_m^{3/2}$ budget	$9\pi/4$	6
Leading geometric-envelope coefficient	3.6831685524	1.8738557776
Thin-tail estimate	$2 + 6(m-1)\rho$	$2 + 3(m-1)\rho$
Additive allowance	10	$5 + \sqrt{2}$
Asymptotic order	$O(n^{1/3} \log n)$	Unchanged

The 49.1238% reduction applies to the displayed leading geometric coefficient, not to the whole upper bound or to observed chain lengths. Earlier selected exact strict examples had $B \approx 1.181996$ at degree 16 and $B \approx 1.001675$ at degree 40. They validated the machinery; they were not claimed to maximize B. A separate degree-421 lower example above 2.6528788584 uses repeated roots and does not address this strict class.

The preceding campaign also found why several shortcuts were insufficient: generic nested hulls can obstruct stronger moment exponents, a scalar geometry/variance relaxation still permits the existing rate, and a logarithmic loss in the generic harmonic lemma is real. None of these is an asymptotic lower bound for actual strict derivative chains. They motivated a campaign aimed at falsifiable local lemmas rather than more constant tuning.

8.3 How fcmaes and experiment-craft were used

The new objective is a nonsmooth numerical function: it contains eigenvalue computations, convexity rejection, minima over paths and maxima over starting roots. Python `fcmaes` 2.1.0 there-

fore fits the discovery layer. BiteOpt and CMA-ES were called in the same process as the evaluator; random search supplied the baseline. A compressed differentiation matrix generated the numerical derivative rows. Curved-line, elliptical-arc, endpoint-cluster and free-coordinate families explored different geometry.

The `optimization-run` command runner was built with Rust 1.88.0. experiment-craft orchestrated frozen inputs, native attempts, dependent certificate and replay steps, declared outputs and reports. Searches ran locally and sequentially, with one numerical thread per search. The application used **trusted command jobs**, not a custom Rust solver worker, a remote fleet or a model call for each evaluation.

Layer	Implementation used	What it establishes
Discovery	Python fcmaes, NumPy, SciPy, numerical geometry and path DP	Candidate parameters and numerical scores
Root proposals	python-flint/Arb	Candidate isolating disks
Exact checker	Gaussian-rational polynomial algebra, Rouché tests, strict orientations and interval DP	The finite statement for the encoded polynomial
Fresh replay	Python with `-S`, using standard-library arithmetic	Rechecks without the optimizer, FLINT or numerical array libraries
Execution evidence	Native command records and sealed outputs, coordinated by experiment-craft	Attempt identity, budgets, byte identity and recorded execution

Promotion does not pretend that floating roots are an exact polynomial. Their recorded decimal coordinates define a new rational input; the code subtracts its **exact** mean, rescales using an outward rational radius and forms its coefficients exactly. The checker then isolates every derivative root of that coefficient-defined polynomial afresh. A promoted certificate checks root completeness, disjointness, all-level strictness, unit-disk containment, the exact centroid and two-sided shortest-path bounds.

This follows the search, check and replay discipline of the CAP application, but it is a **new application-specific checker**. The scalar/PDE CAP audit adapter does not recognize this polynomial application. Its results are ordinary command artifacts, not dashboard `reported_accepted` CAP claims. Only the command runner was built for this campaign; no commissioning of the full three-worker baseline is claimed.

8.4 The campaign as it ran

The first comparison was fixed before its outcomes: random, BiteOpt and CMA-ES; seeds 17 and 29; curve and arc families; degrees 8 and 16; 200 scientific evaluations per arm, including the initial point and repeated points. Later blocks declared their new seeds, degrees and targets before execution. Invalid candidates remained in the evaluation histories. Callback requests beyond the scientific cap returned a rejection score without evaluating another polynomial.

Iterations	Executed work	Decision after the fifth step
91–95	Restore and replay the baseline; validate DP against exhaustive paths; run native search/certify/replay; freeze the comparison	Treat the first exact potential failure as a research lead
96–100	Execute 24 equal-budget arms; certify the stronger degree-8 and degree-16 arc examples	Expand geometry and separate moment and lemma objectives
101–105	Repair signed decimal output; search moments through degree 32; attack endpoint clusters; derive a degree-5 obstruction	Abandon all fixed nonnegative radial coefficients
106–110	Fit a signed potential; attack held-out cases; implement and refute a two-step repair	Stop inflating failed coefficients
111–115	Search chains through degree 48; certify selected examples; implement exact hull moments; find small-degree moment counterexamples	Freeze supported mathematical conclusions
116–120	Reproduce trajectories; replay every certificate; run tests; audit native evidence and history; document failures	Release a frozen evidence archive and fresh-run instructions

The completed campaign contains **96 native jobs in 24 campaigns**: 65 search jobs, 14 certificate jobs, 16 replay jobs and one coefficient fit. The search accounting is **12,024 discovery evaluations plus 800 separately counted repeat evaluations**. All 175 sealed outputs passed the byte-identity audit. There were 45 blocked over-budget callbacks and no caught numerical-evaluation exceptions; infeasible geometries still occurred and remain recorded.

The campaign's suite passed **141 tests**. Four deliberate native repeats reproduced all 800 scientific evaluation records exactly in that environment, excluding timing fields. A separate `python -S` audit replayed all **14 promoted certificates**, matching every stored rational interval requested for those witnesses. Here "independent replay" means independent of the numerical proposal and optimizer dependencies; it is not a second, independently written verifier or a proof-assistant kernel.

8.5 Stronger finite examples

Each row below concerns one specified polynomial, not a supremum over its degree.

Family and degree	Certified lower bound for $B(f)$	Certified upper bound
Arc, 8	1.535182113114249446	1.535182113114249447
Arc, 16	1.712540654859401688	1.712540654859401689
Clustered, 17	1.784968583232396345	1.784968583232396346
Arc, 32	1.836600941824469776	1.836600941824469777
Arc, 48	1.887581747240161414	1.887581747240161415

These are harder strict examples than the earlier selected witnesses. They improve the experimental lower examples and the evidence available to test a proposed upper-bound lemma. They do not improve the universal upper bound, establish growth with degree or prove optimality among strict polynomials.

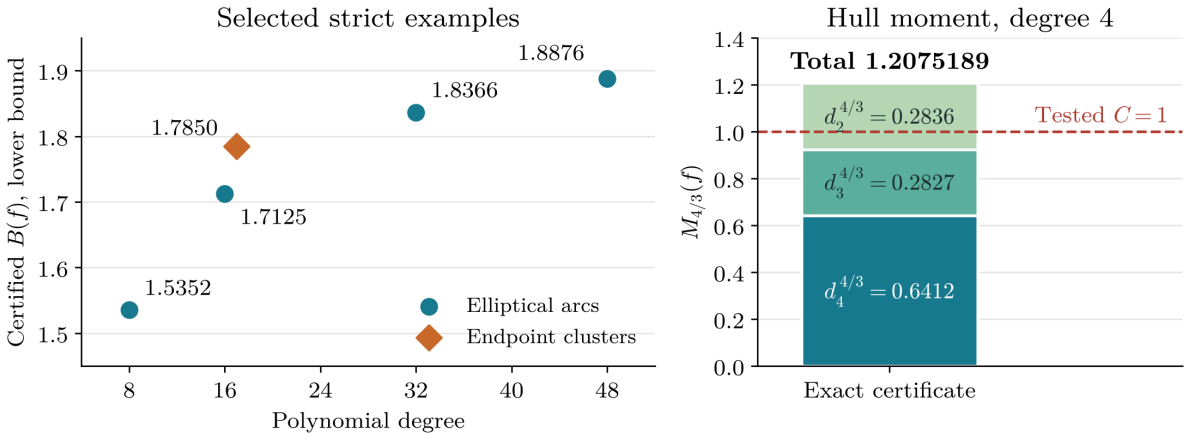


Figure 1: Selected exact chain examples and the exact degree-four hull-moment counterexample. These are finite witnesses; no asymptotic curve has been fitted.

The initial optimizer comparison is also deliberately modest. The following are the best numerical scores over the two predeclared seeds, each with the same 200-evaluation cap:

Arc degree	Random	BiteOpt	CMA-ES
8	1.482084	1.534629	1.535182
16	1.645973	1.712541	1.711259

This illustrates why adaptive search was useful on these particular objectives. Two seeds, selected families and small budgets do not support an overall optimizer ranking. The full per-arm feasibility counts and timings are retained, including the poorer outcomes.

8.6 A failed lemma becomes a mathematical obstruction

A tempting route to a constant bound is a telescoping potential with coefficients a and b,

$$\Phi_m(z) = aR_m + b|z|, \qquad R_m = \max_{v \in Z_m} |v|.$$

If every starting root has a successor satisfying

$$\min_{w \in Z_{m-1}} \left(|z - w| + \Phi_{m-1}(w) - \Phi_m(z)\right) \le 0,$$

then successive choices give $L \le \Phi_n(z_n)$, since $Z_1 = \{0\}$. For $a, b \ge 0$ this would imply $B(f) \le a + b$ in the normalized class, bypassing the accumulation of transport costs in the existing proof.

The initial test with $a = 2$, $b = 1$ failed exactly. Endpoint-cluster optimization found a degree-17 example where even $a = 700$, $b = 1$ leaves a strictly positive local residual. A single polynomial cannot refute every coefficient: whenever its radius drops from one row to the next, a large enough

a makes its residual negative. The next step was therefore to explain the failure with a family rather than merely increase a.

For small $t > 0$, take the five roots

$$x + i\varepsilon x^2, \quad x \in \{-1, -(1-t), 0, 1-t, 1\}, \qquad \varepsilon = t^2,$$

then center and normalize. The top roots form a strict parabola. Simple-root perturbation calculations show that every lower derivative row is also strict for sufficiently small t; the proof checks this explicitly rather than assuming it from the top row. The limiting polynomial is

$$P_0(z) = z(z^2-1)^2, \qquad Z_4 \longrightarrow \{-1, -1/\sqrt{5}, 1/\sqrt{5}, 1\}.$$

The starting root coming from $x = 0$ tends to zero, while R_5 and R_4 both tend to one. Hence, for every fixed $a, b \geq 0$,

$$\min_{w \in Z_4} \left(|z-w| + a(R_4 - R_5) + b(|w| - |z|)\right) \longrightarrow \frac{1+b}{\sqrt{5}} > 0.$$

This is a written proof that **no fixed nonnegative coefficient pair repairs that universal local lemma**, already in degree five. The family member that refutes a given pair must be closer to the limit the larger a is. The repeated-root limit is not the counterexample itself; its nearby, strictly admissible family is. Three finite members were exactly certified. The full expansion and strictness argument are in the archive's `docs/focused/RADIAL_OBSTRUCTION.md`.

Two further repairs were tested:

Proposed repair	Outcome
Fit signed coefficients on 38 selected training examples, then freeze $a = 10.3$, $b = -1.25$	A held-out strict degree-5 example has an exact residual between 0.543181592230508337 and 0.543181592230508338
Allow two actual derivative steps before charging $aR + b\|z\|$, with $a = 100$, $b = 1$	A strict degree-17 example has an exact two-step residual between 0.216896222909042720 and 0.216896222909042721

For a negative b, the candidate unit-disk bound is a, not $a + b$. The two-step checker minimizes over the intermediate root; it does not skip a derivative row. Both positive intervals refute the stated repairs. They do not rule out all signed coefficients, all block lengths or a genuinely different potential.

The conceptual lesson is specific: a prescribed interior starting root may need to move while persistent extreme clusters leave the radius almost unchanged. A useful new invariant must see more than the extreme radius and the starting point's modulus.

8.7 What the hull-moment search established

A uniform $M_{4/3}(f) = O(1)$ budget remains a possible route to a stronger proof. Finding no large values at moderate degrees would not establish it. In fact, after the larger-degree arc searches, small-degree stress tests refuted the newly tested constant one:

$$1.2075188623558127 43 \leq M_{4/3}(f_4) \leq 1.2075188623558127 44.$$

This degree-four polynomial has a complete strict certificate. Any uniform constant valid for this class must therefore be at least the displayed lower bound. It is not a counterexample to the existence of some larger uniform constant.

A simple analytic cubic explains why $C = 1$ was too optimistic. Its centered roots are

$$z_{\pm} = \pm\sqrt{\frac{31}{40}} + \frac{3i}{\sqrt{40}}, \qquad z_0 = -\frac{6i}{\sqrt{40}}.$$

The derivative roots are $\pm 1/\sqrt{30}$ and then zero. All rows are strict, the two outer roots have modulus one, and

$$d_3 = \frac{3}{\sqrt{10}}, \qquad d_2 = \frac{1}{\sqrt{30}}, \qquad M_{4/3} = \frac{10}{30^{2/3}} = \left(\frac{10}{9}\right)^{1/3} > 1.$$

The cubic's worst-start chain is relatively short: from z_0 it has length $\sqrt{14/15} + 1/\sqrt{30} \approx 1.149$. Maximizing the hull moment and maximizing $B(f)$ should therefore remain separate experiments. The elementary proof is in the archive's `docs/focused/TRIANGLE_MOMENT.md`.

The finite moment checker uses exact distances from rational disk centers to convex polygons. If the two root rows have maximum disk radii η_m and η_{m-1}, perturbing the hulls changes their Hausdorff distance by at most $\eta_m + \eta_{m-1}$. Rational outward square roots enclose each distance, and outward cube roots enclose its $4/3$ power. The moment certificate therefore does not rely on floating-point exponentiation.

8.8 Failures that remain part of the result

The inherited tests at first lacked an optional dependency; the failed log was retained, and installing it resolved the failure. A later check found an inherited decimal-formatting error for **negative** rational bounds. Iteration 101 fixed the formatting and added signed outward-rounding tests. Rational intervals, signs, mathematical verdicts and positive chain bounds were unaffected. Early native artifacts keep their original decimal text; the final replay audit provides corrected views.

The release audit also found an operational limitation: **18 of the 24 experiment-craft SQLite workspaces reopened at earlier states despite explicit checkpointing**. The cause in that execution environment is unconfirmed, and the campaign does not claim to have fixed it. A successful single-workspace reopen at iteration 94 did not establish general persistence reliability. To keep a workspace, pause it and use `backup`, as operations describes; a copy of an open database's files can miss its write-ahead log.

All 96 native job records are terminal, all 175 sealed output hashes match, and all 14 mathematical certificates replay. Native journals, frozen inputs, final JSON exports, the mismatched databases and the discrepancy list are preserved. **Do not resume the archived workspaces or automatically resubmit their jobs.** Use their existing native identities for inspection and a fresh campaign for an intentional new experiment. This is a frozen evidence release, not a portable live checkpoint.

8.9 Reproduce, inspect and continue

From the extracted campaign archive, the finite proof audit needs only Python:

```
python -S campaign/replay_all.py --output fresh-exact-audit.json
python -S campaign/audit.py --require-complete --output fresh-audit.json
```

The first command reruns the mathematical checks; the second checks provenance, counts and output identities. With the archive's Git bundle restored, adding `--history` also checks per-iteration evidence hashes against the source snapshots. The final log is `LOG_FOCUSED.md`; the complete current rational results are in `results/focused/117-independent-replay.json`.

For numerical reproduction, install the pinned Python environment and the native runner as the archive's `README.md` describes. The fresh-run script repeats the four arms audited in iteration 116:

```
OPENBLAS_NUM_THREADS=1 OMP_NUM_THREADS=1 \
  .venv/bin/python campaign/reproduce_searches.py \
  --name my-fresh-reproduction
```

The campaign used experiment-craft commit `92c32e0ca90766cdc6df34e8c53fa86496b3c5dc`, optimization-jobs commit `051435c325a51cc89be4ae26645d6937eed6f5e5`, Python fcmaes 2.1.0, NumPy 2.3.5, SciPy 1.17.0, python-flint 0.9.0 and Python 3.12.14. The separate fcmaes-rust 0.1.5 binding supports inherited tests; it was not the optimizer used in these new searches.

fcmaes helped locate adversarial geometries that simple sampling missed; exact checking turned selected candidates into finite mathematical statements; experiment-craft supplied native attempts, frozen inputs and an inspectable evidence trail. Their combination also made failure visible: a training fit could be attacked on held-out inputs, a proposed repair could be refuted, and an operational state discrepancy could be separated from valid sealed mathematical evidence.

The next mathematical target should be a **start-sensitive, derivative-specific invariant**, perhaps one that pays explicitly for escaping persistent clusters and then controls a compatible path. Such a lemma must be stated before searching for its violations. The current campaign supplies adversaries and checks with which to test it. It does not establish an exponent below $1/3$, remove $\log n$, prove a uniform $M_{4/3}$ budget or give a calibrated probability of obtaining those results.

9 Operate experiments through the dashboard

Attach existing workspaces after initialization. For example:

```
.venv/bin/experiment-craft dashboard \
  --workspace "$PWD/guide-runs/scalar/campaign" \
  --workspace "$PWD/guide-runs/pde/campaign" \
  --state-dir "$PWD/guide-dashboard-history"
```

Open the printed temporary-token URL on the coordinator. The service binds only to `127.0.0.1`; there is no public hosting or LAN flag. A physical phone needs a separately secured access path, which this guide does not set up. The token is removed from the address bar after it enters tab storage. Do not publish a token or an active token URL in a screenshot.

9.1 Read the screen in two layers

Start/Continue advances existing work. Pause prevents further dispatch after an already committed attempt; Stop requests native cancellation and its outcome must still be observed. Reconcile & unpause verifies original handles and does not itself start pending work. Unknown attempts are observed, never automatically replayed. A dashboard restart starts no work on its own. The illustrated operator manual shows these controls.

Node charts show sampled saved step counts and observation times. They do not show CPU/RAM/GPU utilization, solver convergence, within-job percentage, ETA or exact historical

execution. Multiple local node roots on one computer remain one physical machine. An accepted step passed its declared execution policy; that alone is not a mathematical theorem.

Use **Load artifacts**, then **Collect** a sealed output and **Download** a checked retained file. For a generated CAP campaign, select the acceptance step's `acceptance` output and press **View audit**. JSON, Markdown and CSV report downloads match the CLI renderer. Every arm stays visible, with its verification/replay outcome, censoring and accounted cost.

Collection follows the advancement lane and is disabled when advancement observations are stale/failed or progress history is unavailable. During a collection, audit reads/downloads wait. Pause/Stop have a separate control lane. Catalog and audit reads are explicit, not repeated on every status poll. After a restarted dashboard stops reporting an operation, refresh artifacts to inspect its actual retained result.

Report status	Read it as
`reported_accepted`	The application's bound receipt says accepted and its native exit/state supports that report
`reported_rejected`	The application reports rejection under its documented exit convention; inspect every arm
`unconfirmed`	A receipt exists, but its native execution/exit does not support the scientific report
`unavailable`	No terminal receipt is recorded; counts/costs remain unknown

`verification-report` validates an admitted application's records. It does not rerun Julia, prove the theorem, or independently trust an external certificate. For untrusted incoming proof data, run the trusted checker and fresh replay before relying on the result. Hashes are not signatures.

The screenshots at the end show the audit of a real scalar campaign, not a mockup or a fixture-copying test job.

10 Prompting agents effectively

Give the agent a mathematical problem card and a precise operational scope. Tell it which evidence would count as success. Ask for immutable problem inputs, a small deterministic test, explicit budgets and a reproducible handover. Keep the evaluator and optimizer together inside one program; do not send one remote/model request per objective evaluation.

10.1 A reusable problem card

```
Question and exact intended conclusion:
Domain, variables, units and representation:
Objective(s), constraints and feasibility convention:
Available derivatives or evaluation code:
Exact constants and physical accuracy target:
What is allowed to vary, and what must stay frozen:
Candidate checker / theorem assumptions:
Evaluation, wall-time, core and memory budgets:
Seeds/starts, baseline and held-out cases:
Required artifacts and independent recheck:
```

Bounds must be mathematically justified, not arbitrary placeholders. For structured variables,

define encoding and repair. A failed simulation needs an explicit outcome; do not turn it into a favorable objective.

10.2 Prompt 1: reproduce and inspect the existing CAP pilot

```
Use the checked experiment-craft checkout and its baseline-lock.json.
Read AGENTS.md, docs/verification-reports.md and the fcmaes-rust
validated-numerics README, THEOREM, SEARCH and AUDIT guides.

Reproduce scalar-cuberoot with the pinned Julia environment on a
fresh one-slot local foreground node. Keep its exact equation,
physical target 1/10000000000, theorem, coefficient construction,
search protocol and verification/replay budgets unchanged.
Run all nine final candidates through fresh verification/replay.
Collect the bound acceptance receipt and produce JSON/CSV/Markdown.
Check the rational bounds separately as the example does.

Report every arm and every failure. Distinguish floating score,
rigorous outcome, native execution and dashboard-reported verdict.
Confirm continuation creates no extra jobs. Return source/binary
hashes, commands, evidence, limitations and an implementation log.
Do not infer optimizer superiority or regional coverage.
```

10.3 Prompt 2: investigate a new conjecture or proof-parameter search

```
First write the problem card for the attached conjecture. Identify
whether the task is candidate search, counterexample search, exact
finite verification, or a theorem-specific interval argument.
Read the shared optimization-jobs ai-context.md and the chosen
solver/application guide. Discover the real model/API before coding.

Build a small independent checker before a long search. Proposals
may be floating point; a claimed counterexample must be checked in
the specified exact or rigorously bounded representation. If the
existing CAP loader does not support this problem, say so and add
an explicit application adapter rather than changing a frozen pilot.

Use a short bounded exploratory campaign and a fixed baseline.
Keep mathematical constants, assumptions and success criteria
separate from tunable search settings. Retain failed/uncertain runs.
A search with no counterexample is not a proof. Return reproducible
inputs, all outputs, checker results and precisely scoped conclusions.
```

10.4 Prompt 3: choose LP/MIP or SQP from the formulation

```
Preserve the required mathematical evidence when choosing a solver.
For a linear model, read LP-IP-craft's ai-context.md, declare all
rows/senses/variable kinds, give every variable of a MIP finite
bounds (continuous ones included), and independently check
objective, row residuals, integrality and any bound/gap report.

For a smooth nonlinear model, read SQP-craft's ai-context.md. Check
the 256-variable limit, supply and test gradient/Jacobian conventions,
use declared starts, and retain primal/KKT residuals and termination.
A first-order point is neither a global optimum nor a second-order
certificate, and a failed solve is not a proof of infeasibility.

Start with a known small instance, then a fresh campaign with stated
limits. Use the native numerical claim verbatim. Any exact theorem
claim needs a separate appropriate checker and stated assumptions.
```

10.5 Prompt 4: review evidence independently

```
Review the frozen mathematical statement, domain, norm, exact
constants, theorem hypotheses, decoder, checker and budget policy.
Treat candidate files as untrusted data. Verify their subject and
input links; rerun the trusted verifier/replay for external proof data.
Check unresolved modes or complete finite coverage, as appropriate.

Try wrong branches, corrupted witnesses, missing/altered tails,
changed inputs, partial ledgers and unsuccessful native executions.
Keep inconclusive, rejected, unavailable and unconfirmed distinct.
Do not edit a bound, flag or target to make an attempt pass.
Return the strongest supported conclusion and an explicit list of
trusted components and remaining mathematical/operational gaps.
```

10.6 If the agent uses MCP

`optimization-mcp` is the runtime interface, not a separate theorem prover. Ask the agent to start with `get_guide`, `describe_node` and `describe_worker`; discover actual schemas and supported models before forming a job. Its normal sequence is `put_input`, `validate_job`, `submit_job`, then `wait_job`/`get_job` and `get_result`/`get_artifact`. After an uncertain submission, look the job up with `get_job` before submitting again. `put_input` stores data, such as an LP model, never code. MCP runs optimization jobs only: trusted command jobs are for the node owner through `optimization-run`, and a new executable or nonlinear model needs explicit local setup, a library program or that trusted-command route. Campaign composition and the dashboard remain the experiment-craft CLI's job; do not invent a campaign MCP tool.

These are prompt templates, not claims that an unattended model has proved a new theorem. The guide's examples were run directly, and the case study lists its own jobs and reproduction commands; no separate model-provider evaluation is implied.

11 Compare methods without changing the question

For ordinary solver studies, experiment-craft supports a protocol declared before dispatch. It fixes groups, allowed variation, seeds/starts, metrics and work/time caps. Use `dry-run`, `init`,

`declare-comparison`, `continue`, then `compare`; the comparison guide gives schemas and a tested 17-job example. All planned observations remain in the denominator. Some of that example's failures are deliberate and its expected campaign status is `incomplete`.

A complete feasible cohort and an available feasible point are different. Unknown work is not zero work. Means/medians are descriptive; the client does not generate a ranking, p-value, confidence interval or optimality theorem. Declared slots and slot-seconds are admission reservations, not measured CPU use. Fixed work and fixed time answer different questions.

CAP audits report final-candidate costs: shared search startup, arm search and fresh verification/replay under the application's accounting. They do not certify intermediate search proposals or measure time to the first certificate found. Failed arms can be informatively censored. There is no successful-only median, survival estimate or proof-aware comparative cohort adapter. Harder/held-out problems, more repetitions and a predeclared anytime certification policy remain future scientific work.

12 Handover, recovery and publication

Keep the frozen mathematical specification, source commits, binary/environment identities, actual candidates, verification/replay records, search/audit protocols, ledgers and checked result files. The campaign retains admitted input snapshots; a report is a convenient view, not a substitute for the workspace and the proof data.

Use maintenance pause and the documented backup and paused restore. Copying only an active SQLite main file can omit its WAL. Maintain one writable campaign authority. An unknown start is reconciled through its original native handle, not rerun because a terminal or agent disconnected.

The useful claim in a paper is specific: what equation/domain/norm was checked, what complete finite set was covered, which assumptions and implementation were trusted, which exact evidence identifies the result, and which failures were retained. A screenshot illustrates operation; machine-readable evidence and a reproducible checker support the scientific statement.

RT-07/RT-08 resource/backend commissioning and automatic placement remain open. The runtime has explicit remote capabilities, but this Python campaign client does not route its campaigns to SSH workers. A prepared inventory with six disabled workers is not a six-machine validation. Independently commissioned hosts, secured phone access, GPU/resource isolation and portable semantic replay require separate evidence.

13 Sources and version boundary

Inspected source	Revision
experiment-craft	`63235813cbd2906d01792f39d484e0c138717804`
optimization-jobs	`051435c325a51cc89be4ae26645d6937eed6f5e5`
fcmaes-rust	`c5a83e21484892c2c72b4c5c282ae72d1c44b753`
LP-IP-craft	`e5cf5d04322a0ddecfb364b392ef92288bd27ee2`
SQP-craft	`687b78e22d41e3843226398191c8c62c0246a998`
fast-cma-es (optional Python route)	`3ba89a7314165764a6e7b126ddc0711cb79cf363`
pykep-core-rust (optional physics library)	`542158b575ad9d235841d501772c179177d69660`

The four native revisions are those of `baseline-lock.json`. The optional Python and astrodynamics libraries are described from their sources at these revisions; they are not part of the orchestrator's locked native baseline. The derivative-root campaign ran at the later revisions given in its reproduction section.

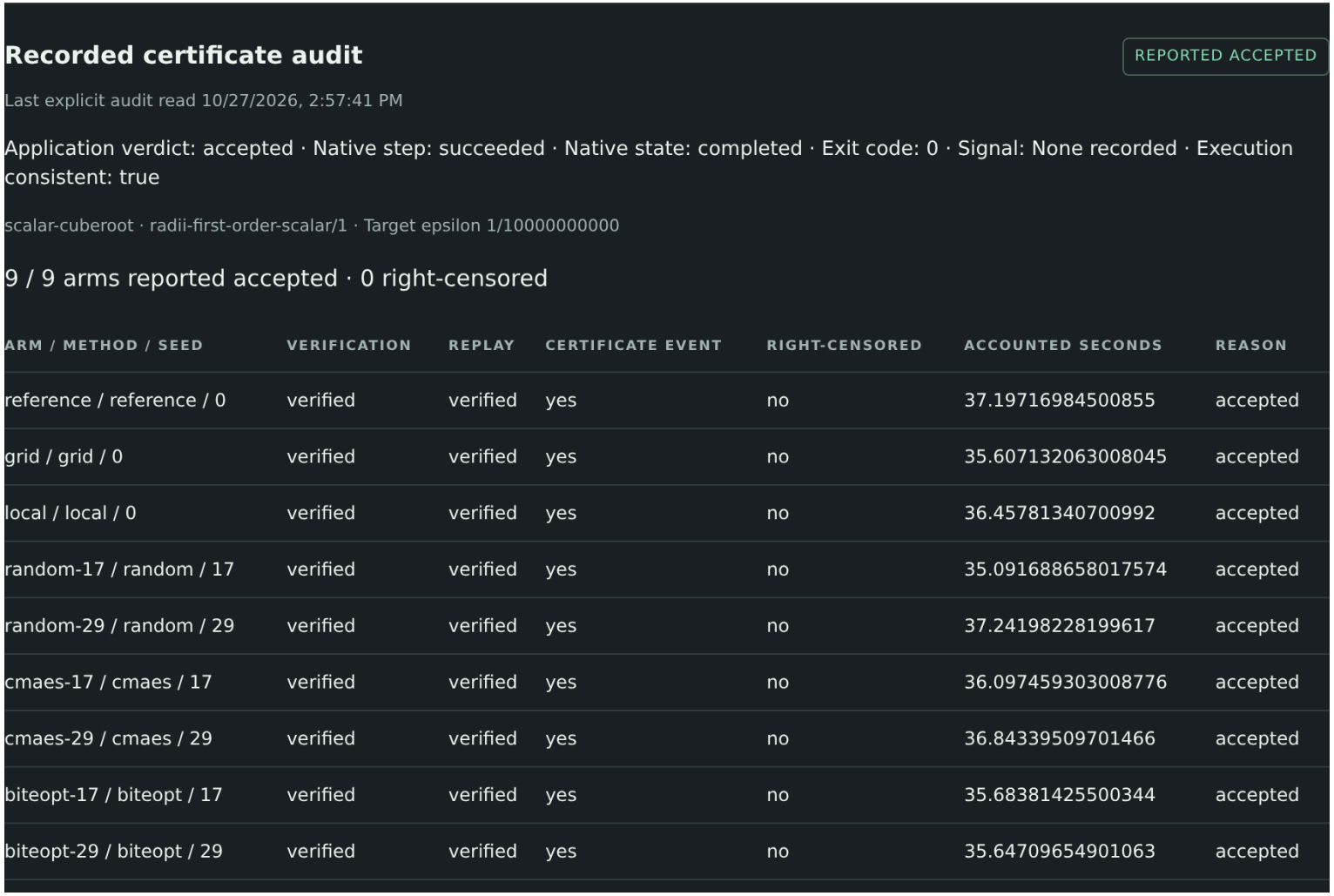
Read next:

- Shared solver and agent guide: interfaces, modeling, native evidence and limits.
- CAP mathematical contract, bounded search and all-arm audit.
- Artifacts and report semantics, dashboard reference and illustrated operator manual.
- LP/IP agent guide and SQP agent guide.
- Prompt-driven application tutorials, CFD prompt workshop and astrodynamics scope.

The scalar and Cahn-Hilliard formulas are adapted from RadiiPolynomial's examples, as the application's README credits; it retains their MIT notice in `RADII_POLYNOMIAL_LICENSE.txt`. This guide claims no new attribution or independent theorem implementation.

14 Screenshots of a real scalar audit

Both images were captured from the dashboard reading a scalar campaign that ran the genuine verifier and replay; its receipt was not copied from a test fixture. They omit the address bar and access token. The PDF places the wide view on a landscape page.



Recorded certificate audit

REPORTED ACCEPTED

Last explicit audit read 10/27/2026, 2:57:41 PM

Application verdict: accepted · Native step: succeeded · Native state: completed · Exit code: 0 · Signal: None recorded · Execution consistent: true

scalar-cuberoot · radii-first-order-scalar/1 · Target epsilon 1/10000000000

9 / 9 arms reported accepted · 0 right-censored

ARM / METHOD / SEED	VERIFICATION	REPLAY	CERTIFICATE EVENT	RIGHT-CENSORED	ACCOUNTED SECONDS	REASON
reference / reference / 0	verified	verified	yes	no	37.19716984500855	accepted
grid / grid / 0	verified	verified	yes	no	35.607132063008045	accepted
local / local / 0	verified	verified	yes	no	36.45781340700992	accepted
random-17 / random / 17	verified	verified	yes	no	35.091688658017574	accepted
random-29 / random / 29	verified	verified	yes	no	37.24198228199617	accepted
cmaes-17 / cmaes / 17	verified	verified	yes	no	36.097459303008776	accepted
cmaes-29 / cmaes / 29	verified	verified	yes	no	36.84339509701466	accepted
biteopt-17 / biteopt / 17	verified	verified	yes	no	35.68381425500344	accepted
biteopt-29 / biteopt / 29	verified	verified	yes	no	35.64709654901063	accepted

Figure 2: Recorded positive-root audit of a real scalar campaign: native execution and the reported verdict are separate, and all nine final arms remain visible. This view runs no checker.



Figure 3: Phone-sized view of the same audit. The arm table scrolls sideways. This is Chromium touch emulation, not a physical-phone or remote-access commissioning result.